

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1433

Waterford Power Llc 500 kV
April 20, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1433. The Interconnection Customer proposes a 197.6 MW solar photovoltaic facility interconnecting at the Waterford Power Llc 500 kV substation in the Ohio area. The request was received on March 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 522 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$52,553,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carbon Ctr 500/345 kV TX
Contingency: Blackrok to Shawville Station 345 kV
Loading: 122.2% (Rating: 1,998 MVA)
DFAX: 6.0%
Pre-project loading: 121.6%
2. Facility: Carbon Ctr to Shawville Station Substation 500 kV
Contingency: Blackrock to Shawville Station 345 kV L
Loading: 115.1% (Rating: 2,111 MVA)
DFAX: 6.0%
Pre-project loading: 114.6%
3. Facility: Orchard Road to Shawville Station 500 kV
Contingency: Shawville Sta to Scott 345 kV
Loading: 111.4% (Rating: 2,944 MVA)
DFAX: 7.7%
Pre-project loading: 110.8%
4. Facility: Oreg 3 Inc 345-230 kV Transformer
Contingency: Tidd to Muskingum River 500 kV
Loading: 106.2% (Rating: 310 MVA)
DFAX: 6.8%
Pre-project loading: 101.9%
5. Facility: Brentwood to Muskingum Riv 230 kV
Contingency: Tidd to Muskingum Riv 500 kV
Loading: 104.4% (Rating: 580 MVA)
DFAX: 9.8%
Pre-project loading: 101.1%

6. Facility: Mine Rd to Brentwod 230 kV line
 Contingency: Muskingum River to Tidd 500 kV
 Loading: 103.9% (Rating: 573 MVA)
 DFAX: 9.8%
 Pre-project loading: 100.5%
7. Facility: Mine Oad to West Bellaire 230 kV line
 Contingency: Tidd to Muskingum Riv 500 kV
 Loading: 103.7% (Rating: 566 MVA)
 DFAX: 9.8%
 Pre-project loading: 100.3%
8. Facility: Ft Humbug to Coronet 500 kV
 Contingency: Montour Sta to Acadia 500 kV
 Loading: 100.6% (Rating: 3,346 MVA)
 DFAX: 8.2%
 Pre-project loading: 100.1%
9. Facility: Blackrock to Shawville Station 345 kV
 Contingency: Carbon Ctr Substation 500-345 kV Autotransformer
 Loading: 100.5% (Rating: 2,489 MVA)
 DFAX: 7.0%
 Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carbon Ctr 500/345 kV TX	Replace transformer with higher-rated unit	\$28,319,000	2% (\$663,000)
Carbon Center to Shawville Station 500 kV L	Rebuild line with higher-capacity conductor	\$106,341,000	2% (\$2,491,000)
Orchard Road to Shawville Station 500 kV	Rebuild line with higher-capacity conductor	\$239,279,000	2% (\$4,773,000)
Oreg 3 Inc 345-230 kV Transformer	Replace transformer with higher-rated unit	\$28,053,000	69% (\$19,423,000)
Brentwood to Muskingum Riv 230 kV	Reconductor line	\$19,352,000	54% (\$10,505,000)
Brentwood to Mine Road 230 kV	Reconductor line	\$10,858,000	54% (\$5,894,000)
Mine Oad to West Bellaire 230 kV line	Reconductor line	\$14,021,000	54% (\$7,612,000)
Ft Humbug to Coronet 500 kV	Reconductor line	\$40,823,000	2% (\$751,000)
Blackrock to Shawville Sta 345 kV	Reconductor line	\$21,801,000	2% (\$441,000)
	Total allocated to project		\$52,553,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.